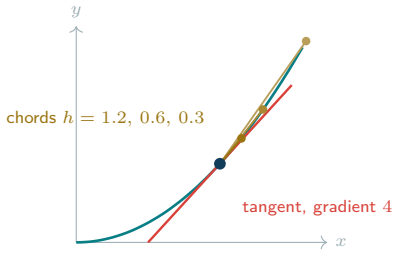
SEQUENCES LADDER 7 OF 9

From differences to the derivative

Shrinking the step in a difference quotient to zero turns the gradient of a chord into the gradient of the curve.

1 / THE RULE AND THE PICTURE

The difference quotient with a smaller and smaller step h gives the derivative



For a rule $f(x)$ the first difference with step h divided by h is the gradient of the chord from x to $x+h$, and as the step shrinks the chords settle on the tangent:

$$f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$$

Up to now h was 1, so the first difference of a sequence is the gradient of a chord of step 1.

2 / WATCHING THE GRADIENTS SETTLE

Three examples with smaller and smaller steps

step h	x^2 at $x=3$	x^3 at $x=2$	$5t^2$ at $t=2$
1	7	19	25
0.1	6.1	12.61	20.5
0.01	6.01	12.0601	20.05
0.001	6.001	12.006001	20.005

For x^2 at $x=3$ the quotient is $6+h$, for x^3 at $x=2$ it is $12+6h+h^2$, and for the distance $s=5t^2$ metres the average speed over $[2,2+h]$ is $20+5h$ metres per second, so the speed at $t=2$ is 20 m/s, matching $v=10t$.

3 / THE POWER RULE FROM THE BINOMIAL THEOREM

The quotient is a polynomial in h whose first term survives

Expanding $(x+h)^n = x^n + nx^{n-1}h + \binom{n}{2}x^{n-2}h^2 + \cdots$ gives $\frac{(x+h)^n-x^n}{h} = nx^{n-1} + \binom{n}{2}x^{n-2}h + \cdots$, and every term except the first has a factor h and vanishes as $h\to 0$, so $\frac{d}{dx}x^n = nx^{n-1}$. With $h=1$ the same expansion gives the whole difference, $\Delta n^3 = 3n^2+3n+1$, whose first term $3n^2$ is the derivative.

4 / SECOND DIFFERENCES BECOME THE SECOND DERIVATIVE

The change of the gradient

The second difference with step h over h^2, $\frac{f(x+2h)-2f(x+h)+f(x)}{h^2}$, tends to $f''(x)$, and for $f=x^3$ at $x=1$ with $h=0.1$ it is $\frac{1.728-2.662+1}{0.01} = 6.6$, heading to 6.
For a quadratic an^2+bn+c it equals $2a$ for every h, which is why a constant second difference means a constant second derivative.

5 / WHAT CALCULUS ADDS

Every step size, not only whole numbers

Differences only compare terms a whole step apart, while the derivative asks about the instant, by letting the step run through $0.1, 0.01, 0.001$ and beyond.
The rules are the ones met in the earlier sheets, since the power rule is the leading term of the difference rule, and the second difference of a quadratic is its second derivative.

6 / TRY THESE

Find the gradient numerically

(a) For x^4 at $x=1$ with $h=0.01$ the quotient is 4.060401, heading to 4. **(b)** For a car with distance $s=5t^2$ the average speed from $t=1$ to $t=2$ is 15 m/s, and over $[2, 2.1]$ it is 20.5 m/s, so the speed at $t=2$ is 20 m/s. **(c)** Show that for $f=x^2$ the quotient is exactly $2x+h$, so the limit is $2x$.

ASIDE / THE FIRST LESSON

The derivative is a limit of gradients of chords, and the sequences met so far are the chords of step 1.

LINKS TO OTHER SHEETS

The next sheet does the same for sums and areas; the physics hinge sheets use the derivative as a rate.

Convention. h is the step; $f'(x)$ the derivative; $\Delta f = f(x+h)-f(x)$; the chord gradient is $\Delta f/h$.